Lab Programs: Week 4

1. Write the JavaScript Code to generate a multiplication table.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Multiplication Table</title>
</head>
<body>
    <script>
        let n=prompt("enter n:")
        for(let i=1;i<=10;i++){
            document.write(`${n}x${i}=${n*i}<br>`)
        }
    </script>
</body>
</html>
```

Output:

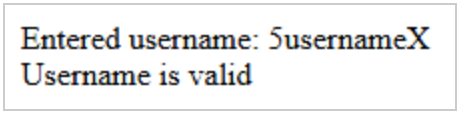
5x1=5
5x2=10
5x3=15
5x4=20
5x5=25
5x6=30
5x7=35
5x8=40
5x9=45
5x10=50

2. Write JavaScript code to validate a 10-character username that starts with a digit and ends with an alphabet.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Validate username</title>
</head>
<body>
    <script>
        let username=prompt("enter your username:")
        let pattern=/^\d[A-Za-z]{8}[A-Za-z]$/
        if (pattern.test(username)){
            document.write("Username is valid")
        }
        else{
            document.write("Username is invalid")
        }
    </script>
</body>
</html>
```

Output:



3. Write JavaScript code to move an image in the direction selected by the user using Left, Right, Up, and Down buttons.

Code:

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Image Moving using buttons</title>
    <style>
      .buttons {
        display: flex;
        margin: auto;
        justify-content: center;
      }
      body {
        margin: 20px;
        padding: 30px;
      }
      img {
        position: absolute;
        left: 150px;
        top: 150px;
      }
    </style>
  </head>
  <body>
    <img
      class="image"
      src="https://png.pngtree.com/png-vector/20220121/ourmid/pngtree-3d-tennis-ball-vector-ilustration-
png-image_4331248.png"
      alt="tennisball"
      width="150px"
      height="150px"
    />
    <br /><br /><br /><br /><br />
    <div class="buttons">
      <button onclick="moveLeft()">Left</button>
      <button onclick="moveRight()">Right</button>
      <button onclick="moveDown()">Down</button>
      <button onclick="moveUp()">Up</button>
    </div>
    <script>
      let img = document.querySelector(".image");
      let x = 150;
      let y = 150;
      function moveLeft() {
        x = x - 10;
        img.style.left = x + "px";
      }
      function moveRight() {
        x = x + 10;
        img.style.left = x + "px";
      }
      function moveUp() {
        y = y - 10;
        img.style.top = y + "px";
```

```
      }
      function moveDown() {
        y = y + 10;
        img.style.top = y + "px";
      }
    </script>
  </body>
</html>
```

Output:

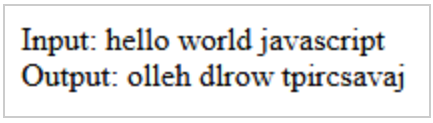


4. Write a JavaScript function that accepts a string as a parameter and reverses each word in the string.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Reverse a string</title>
</head>
<body>
    <script>
        let word=prompt("enter something: ")
        document.write(word.split(" ").map(word=>word.split("").reverse().join("")).join(" "))
    </script>
</body>
</html>
```

Output:



5. Write a JavaScript function that takes an array of numbers and finds the second smallest and second largest numbers.

Code:

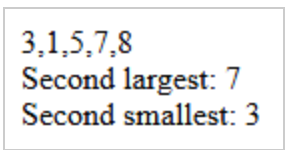
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Find second smallest and second largest</title>
</head>
<body>
    <script>
        let n=[3,1,5,7,8]
       document.write(n+"<br>")

        function secondLargest(arr){
            let max=arr[0];

            for(let i=1;i<arr.length;i++){
                if(arr[i]>max) max=arr[i]
            }
            let second=arr[0]
            for(let i=1;i<arr.length;i++){
                if(arr[i]<max && arr[i]>second) second=arr[i]
            }
            document.write(`Second largest: ${second}`)
        }
        function secondSmallest(arr){
            let max=arr[0];

            for(let i=1;i<arr.length;i++){
                if(arr[i]<max) max=arr[i]
            }
            let second=arr[0]
            for(let i=1;i<arr.length;i++){
                if(arr[i]>max && arr[i]<second) second=arr[i]
            }
            document.write(`Second smallest: ${second}`)
        }
         secondLargest(n)
         document.write("<br>")
        secondSmallest(n)
    </script>
</body>
</html>
```

Output:



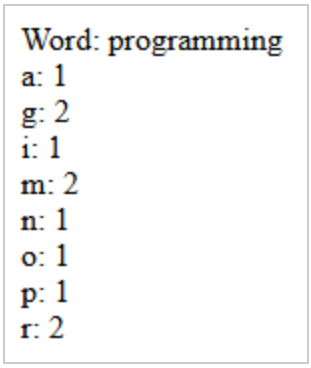
6. Write a JavaScript program that takes a word as input and finds the number of occurrences of each letter in the given String.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Frequency of each character in a word</title>
</head>
<body>
    <script>
        let word=prompt("enter a word")
        let freq=new Array(26).fill(0)

        for(let i=0;i<word.length;i++){
            let index=word[i].charCodeAt(0)-'a'.charCodeAt(0)
            freq[index]++;
        }
        for(let i=0;i<26;i++){
            if(freq[i]>0){
                document.write(`${String.fromCharCode(i+97)}: ${freq[i]}<br>`)
            }
        }
    </script>
</body>
</html>
```

Output:



7. Write a JavaScript function that generates all possible substrings of a given string.

Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>All Substrings</title>
</head>
<body>

<script>
    function substrings(str) {
        for (let i = 0; i < str.length; i++) {
            for (let j = i + 1; j <= str.length; j++) {
                document.write(`${str.substring(i, j)}, `);
            }
        }
    }

    let word = prompt("enter a word: ")

    substrings(word);
</script>

</body>
</html>
```

Output:

Word: hello
h, he, hel, hell, hello, e, el, ell, ello, l, ll, llo, l, lo, o,

8. Write a JavaScript function to find the first not repeated character.

Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>First Non-Repeated Character</title>
</head>
<body>

<script>
    function firstNonRepeated(str) {
        let freq = {};

        for (let ch of str) {
            freq[ch] = (freq[ch] || 0) + 1;
        }

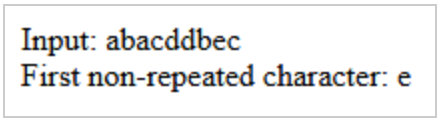
        for (let ch of str) {
            if (freq[ch] == 1) {
                return ch;
            }
        }
    }

    let str = prompt("enter a word: ")

    document.write(`First non-repeated character: ${firstNonRepeated(str)}`);
</script>

</body>
</html>
```

Output:



9. Write a JavaScript program to check the strength of a given password and display whether it is a Strong Password or not.

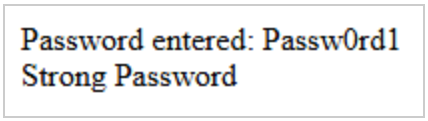
Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Password Strength</title>
</head>
<body>
    <script>
    let password = prompt("Enter your password");

    let pattern = /^(?=.*[A-Z])(?=.*\d).{8,}$/;

    if (pattern.test(password)) {
        document.write(`Strong Password`);
    }
    else {
        document.write(`Not a Strong Password`);
    }
</script>
</body>
</html>
```

Output:



10. Write a JavaScript program to remove duplicate characters from a given string.

Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>Remove Duplicate Characters</title>
</head>
<body>

<script>
    let str = prompt("Enter a string");
    let result = "";

    for (let ch of str) {
        if (!result.includes(ch)) {
            result += ch;
        }
    }

    document.write(`Result: ${result}`);
</script>

</body>
</html>
```

Output:

